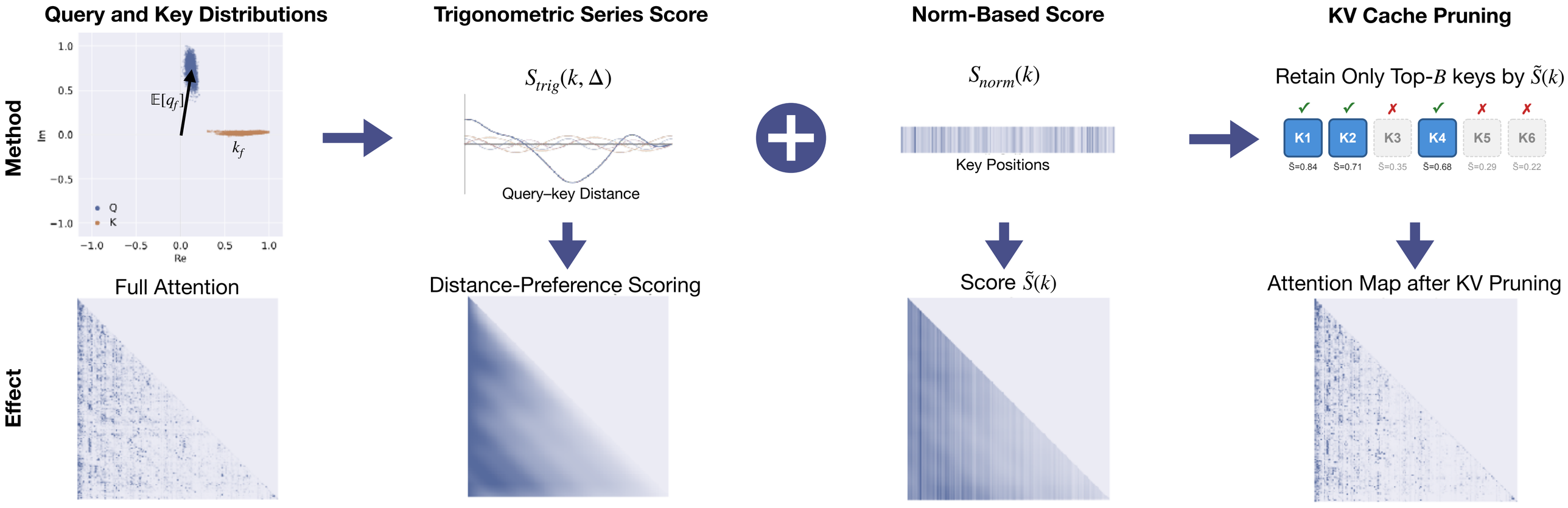
Query and Key Distributions
Trigonometric Series Score
Norm-Based Score
KV Cache Pruning
Method
Effect
1.0
0.5
0.0
−0.5
−1.0
Im
Re
$\mathbb{E}[q_f]$
k_f
Q
K
$S_{trig}(k, \Delta)$
Query–key Distance
$S_{norm}(k)$
Key Positions
Retain Only Top-B keys by $\tilde{S}(k)$
K1
K2
K3
K4
K5
K6
$\tilde{S}$=0.84
$\tilde{S}$=0.71
$\tilde{S}$=0.35
$\tilde{S}$=0.68
$\tilde{S}$=0.29
$\tilde{S}$=0.22
Full Attention
Distance-Preference Scoring
Score $\tilde{S}(k)$
Attention Map after KV Pruning